BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MULTAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

COMPUTER SCIENCE

Past Paper 2024 • Paper Code: 4831 / 4834

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

COMPUTER SCIENCE PAPER-II GROUP-I (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** 4831

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that bubble in front of that question number, on bubble sheet. Use marker or pen to fill the bubbles. Cutting or filling two or more bubbles will result in zero mark in that question.

Q. (Either from "MS-ACCESS & C-LANGUAGE" OR from "MS-ACCESS & VISUAL BASIC")

Q. MS-ACCESS

Q1. A collection of related field is called:

(A) Character
(B) Record
(C) Database
(D) File

Q2. CREATE command is used to create:

(A) Record
(B) View
(C) Report
(D) Query

Q3. Customers, cars are examples of:

(A) Entity
(B) Attribute
(C) Cardinals
(D) Relationship

Q4. The columns of a table corresponds to:

(A) Table
(B) Record
(C) Field
(D) Cell

Q5. In 3NF, a non-key attribute must not depend on a:

(A) Non-key attribute
(B) Key attribute
(C) Composite key
(D) Sort key

Q6. In a relational database, a single piece of information is called:

(A) Table
(B) Record
(C) Entity
(D) Attribute

Q. C-LANGUAGE

Q7. The extension of header file is:

(A) .cpp
(B) .txt
(C) .E
(D) .h

Q8. Variable and constant name can not contain a(n):

(A) Number
(B) Underscore
(C) Letter
(D) Period

Q9. The function that is used to display output on screen is called:

(A) printf
(B) scanf
(C) pow
(D) display

Q10. The format specifier %d is used for:

(A) 81
(B) 'F'
(C) 4.83
(D) '9'

Q11. An other term for a computer making a decision is:

(A) Sequential
(B) Iteration
(C) Selection
(D) Repetition

Q12. When x = 40, f = (x < 50) ? 10 : 20;, after executing the statement, the value of x is:

(A) 20
(B) 10
(C) 40
(D) 100

Q13. A loop within a loop is called:

(A) Nested loop
(B) Complex loop
(C) Infinite loop
(D) Dual loop

Q14. A built in function:

(A) Can not be redefine
(B) Can be redefined
(C) Can not return a value
(D) Should be redefined

Q15. The mode opens only an existing file for both reading and writing is:

(A) "W" Output mode
(B) "W+" Append mode
(C) "R+" Create mode
(D) "A+" Read mode

Q. OR

Q. VISUAL BASIC

Q7. The diamond in a flow-chart indicates:

(A) Progress
(B) Input
(C) Output
(D) Condition

Q8. Variable and constant name can not contain a(n):

(A) Number
(B) Underscore
(C) Letter
(D) Period

Q9. BASIC was introduced in:

(A) 1960 AD
(B) 1950 AD
(C) 1970 AD
(D) 1980 AD

Q10. Which event detects the processing of standard displayable characters?

(A) Key down
(B) Key up
(C) Key press
(D) Keyboard

Q11. An other term for a computer making a decision is:

(A) Sequential
(B) Iteration
(C) Selection
(D) Repetition

Q12. When a relational expression is false, it has the value:

(A) 1
(B) 0
(C) Less than zero
(D) −1

Q13. A loop within a loop is called:

(A) Nested loop
(B) Complex loop
(C) Infinite loop
(D) Dual loop

Q14. A built in function:

(A) Can not be redefine
(B) Can be redefined
(C) Can not return a value
(D) Should be redefined

Q15. To create a new record, the file must be opened in:

(A) Output mode
(B) Append mode
(C) Create mode
(D) Read mode

COMPUTER SCIENCE PAPER-II GROUP-I (Subjective Type)

Time Allowed: 2.10 Hours | **Maximum Marks:** 60

> Note: Write same question number and its parts number on answer book, as given in the question paper.

Q. SECTION-I

Q. (Either from "MS-ACCESS & C-LANGUAGE" OR from "MS-ACCESS & VISUAL BASIC")

Q2. (Marks 6 × 2 = 12) Write short answers to any six parts. MS-ACCESS

(i) What is the main purpose of database system?

(ii) Define foreign key.

(iii) How many fields and records does the following table contain? Roll No. | Name | Subject | Class; 01 | Ali | English | 1st year; 02 | Zia | Math | 1st year

(iv) What is the primary objective of ER diagram?

(v) Differentiate between entity class and entity instance.

(vi) Define partial dependency.

(vii) What is database wizard?

(viii) Write the use of wild cards.

(ix) What is the purpose of radio button?

Q3. (Marks 6 × 2 = 12) Write short answers to any six parts. C-LANGUAGE

(i) Why comments are used in program?

(ii) Write the given expression in C-Language: $a = 2b^2c^3$

(iii) Find the errors in the following code: integer x, y; x=10, y=20.5;

(iv) Determine the output of the following code: int x = 1; x++; printf ("%d", x++);

(v) What is control string in printf ( ) function?

(vi) Find errors in the following code: void main ( ) { printf (OK) }

(vii) What will be the output of following code: printf ("\""Hello \\\ n Dear!");

(viii) Define text stream.

(ix) What does "A+" mean in file handling?

Q. OR

Q3. (Marks 6 × 2 = 12) Write short answers to any six parts. VISUAL BASIC

(i) Describe categorized view of property in properties window.

(ii) Define the use of Command Button Control.

(iii) Differentiate between the hide and show methods of form.

(iv) Define Modeless form.

(v) What is the use of Query Unload Event?

(vi) Define List Index Property.

(vii) What is the purpose of the statement 'Unload Me'?

(viii) How can two groups of option buttons work independently?

(ix) How is information placed inside a combo box?

Q4. (Marks 6 × 2 = 12) Write short answers to any six parts. C-LANGUAGE

(i) Why break statement is used in a switch statement?

(ii) Trace the output of the following code segment: int p = 3, q = 5; if ((p > q) || (p! = 4)) p = p + 1; else p = p - 1; printf ("%d", p);

(iii) Define infinite loop.

(iv) Convert the following code into while ( ) loop: void main (void){ int a, k = 1; for (a = 1; a <= 10; a++) printf ("%d\n", k+a);}

(v) Predict the output: int i, a = 3; for (i = 5; i > 0; i --){ a = a + i; printf ("%d\n", a);}

(vi) What is the life time of a local variable?

(vii) Why return statement is used in function?

(viii) How is a function activated?

(ix) What is the use of formal parameters?

Q. OR

Q4. (Marks 6 × 2 = 12) Write short answers to any six parts. VISUAL BASIC

(i) What is concatenation in Visual Basic?

(ii) What is Val Function?

(iii) Define Infinite loop.

(iv) Differentiate between "+" and "&" in Visual Basic.

(v) How select --- case help in coding.

(vi) Write the syntax of while --- wend loop.

(vii) What is default property of command button?

(viii) Which statements terminate Do and For loops early?

(ix) Trace the output: If (10 Mod 3 = 0) Then Print ("YES") Else Print ("NO") End if

Q. SECTION-II

QNOTE. Attempt any one question. MS-ACCESS

Q5. (Marks 8) What are the database anomalies? Discuss different anomalies in database.

Q6. (Marks 8) Explain any four types of queries in MS-Access.

QNOTE. Attempt any two questions. Either from "C-Language" OR "Visual Basic"

Q. C-Language

Q7. (Marks 8) Define bugs. Explain different types of errors in C-program.

Q8. (Marks 8) Write a program that inputs an integer and determines whether it is positive, negative or zero.

Q9. (Marks 8) Write a program that displays odd numbers from 1 to 17 using any loop.

Q. OR

Q. VISUAL BASIC

Q7. (Marks 8) Define bugs. Explain different types of errors in Visual Basic.

Q8. (Marks 8) Write a program that inputs an integer and determines whether it is positive, negative or zero.

Q9. (Marks 8) Write a program that displays odd numbers from 1 to 17 using any loop.

GROUP – II

COMPUTER SCIENCE PAPER-II GROUP-II (Objective Type)

Time Allowed: 20 Minutes	**Maximum Marks:** 15	**Paper Code:** 4834

> Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that bubble in front of that question number, on bubble sheet. Use marker or pen to fill the bubbles. Cutting or filling two or more bubbles will result in zero mark in that question.

Q. (Either from "MS-ACCESS & C-LANGUAGE" OR from "MS-ACCESS & VISUAL BASIC")

Q. MS-ACCESS

Q1. Which database object is used to retrieve data from database?

(A) Queries
(B) Forms
(C) Reports
(D) Tables

Q2. The data in tables is entered in:

(A) Design view
(B) Normal view
(C) Datasheet view
(D) Layout view

Q3. Which is also known as data set?

(A) Module
(B) File
(C) Record
(D) Field

Q4. A relation is also known as:

(A) Table
(B) Tuple
(C) Attribute
(D) Relationship

Q5. In an ER diagram, a rectangle represents:

(A) Attribute
(B) Field
(C) Flow
(D) Entity

Q6. Different attributes in two different tables having same name are called:

(A) Synonym
(B) Acronym
(C) Homonym
(D) Mutually exclusive

Q. C-LANGUAGE

Q7. A function can return _____ value(s).

(A) 1
(B) 2
(C) 3
(D) Many

Q8. Which function is used to write a string to a file?

(A) puts
(B) printf
(C) fputs( )
(D) getch

Q9. Void occupy how many bytes in memory?

(A) Four
(B) Three
(C) Two
(D) Zero

Q10. For initializing a variable, we use _____ operator.

(A) →
(B) =
(C) ?
(D) @

Q11. The function getch( ) is defined in:

(A) conio.h
(B) stdio.h
(C) math.h
(D) input.h

Q12. If int z = 222, then the statement printf("%5d",z) will display:

(A) □222□
(B) 222□□
(C) □□222
(D) "222"

Q13. What will be the value of n after executing the following statement: n=(100 > 200) ? 100 : 200 ;

(A) 100
(B) 200
(C) 100200
(D) 200100

Q14. Which is valid case statement in a switch?

(A) case x < 2
(B) case "one"
(C) case 1.2
(D) case 1 :

Q15. How many types of looping structure are available in C?

(A) 2
(B) 3
(C) 4
(D) 1

Q. OR

Q. VISUAL BASIC

Q7. The property that indicates the number of selected item in the list box control is:

(A) selcount
(B) count
(C) sum
(D) index

Q8. The property that specifies maximum number of characters that can be entered in a text box:

(A) length
(B) max
(C) max length
(D) multi length

Q9. Forms and controls are collectively recognized as:

(A) Properties
(B) Methods
(C) Events
(D) Object

Q10. Methods characterize an object's:

(A) Appearance
(B) Behaviour
(C) Value
(D) Style

Q11. Variable and constant names can not contain:

(A) Periods
(B) Number
(C) Underscore
(D) Letter

Q12. The integer data type in VB occupies _____ bytes.

(A) 1
(B) 12
(C) 2
(D) 8

Q13. The loop which never ends is called _____ loop.

(A) nested
(B) infinite
(C) inner
(D) outer

Q14. If number of iterations are known in advance, the suitable loop is:

(A) Do loop while
(B) Do loop until
(C) Do while loop
(D) For Next

Q15. Image control holds a graphic that is assigned to _____ property.

(A) caption
(B) picture
(C) text
(D) auto size

COMPUTER SCIENCE PAPER-II GROUP-II (Subjective Type)

Time Allowed: 2.10 Hours | **Maximum Marks:** 60

> Note: Write same question number and its parts number on answer book, as given in the question paper.

Q. SECTION-I

Q. (Either from "MS-ACCESS & C-LANGUAGE" OR from "MS-ACCESS & VISUAL BASIC")

Q2. (Marks 6 × 2 = 12) Write short answers to any six parts. MS-ACCESS

(i) What is meant by Data manipulation?

(ii) What is the purpose of using views?

(iii) How many fields and records are contained in following table? Roll No. | Student name | Class; 1201 | Asad | 2nd year; 1205 | Majid | 2nd year; 1212 | Saima | 1st year

(iv) Why is requirement analysis conducted?

(v) Define entity with example.

(vi) What is meant by data integrity?

(vii) What are major database objects in MS-Access?

(viii) How table design view is different from datasheet view?

(ix) Define different layouts of form.

Q3. (Marks 6 × 2 = 12) Write short answers to any six parts. C-LANGUAGE

(i) Give examples of any two standard identifiers in C-language.

(ii) Convert the express x ≥ y ≥ z in C-language formate.

(iii) What is the output of the following code? int y=10; int x=2; y = x+y; x = x+5; printf("x = %d and y =%d", x, y);

(iv) Indicate the errors in the following lines: int 5=x; x =+x; printf('%s', x);

(v) Clarify the difference between getch( ) and getche( ) functions.

(vi) Find out the errors: void main (void) { int total marks = 500; int 0M = 300; float p = 0M *100 ÷ total marks; printf ("%f", p);

(vii) Predict the output; int x=y=5; printf ("%d \n %d", ++x, x++);

(viii) What is the difference between printf( ) and fprintf( ) functions?

(ix) Write purpose of 'fopen' and 'fclose' functions.

Q. OR

Q3. (Marks 6 × 2 = 12) Write short answers to any six parts. VISUAL BASIC

(i) What is the purpose of 'name' property of a control?

(ii) Name any two components of IDE.

(iii) What is the difference between method and property?

(iv) What is the purpose of control's naming conventions?

(v) Differentiate between list-box and combo-box.

(vi) How can you enter data in multiple lines of a text box?

(vii) What is the purpose of tab index property?

(viii) How can two groups of option buttons work independently?

(ix) What is the purpose of timer control?

Q4. (Marks 6 × 2 = 12) Write short answers to any six parts. C-LANGUAGE

(i) Define compound statement.

(ii) Determine the output of the following code segment: char ch = 'a'; switch (ch) { Case 'a': printf ("A"); case 'b': printf ('B'); }.

(iii) Write two uses of loop.

(iv) Convert the following code segment into while loop. Void main ( ) { int i; for (i = 1; i <= 20; i++) printf ("loop") ; }.

(v) Predict the output of the following code segment. int n; for (n=7; n>=1; n--) printf ("%d", n) ;

(vi) Define function header.

(vii) How does a function return a value?

(viii) Write down the use of actual parameter.

(ix) Why is function used in a program?

Q. OR

Q4. (Marks 6 × 2 = 12) Write short answers to any six parts. VISUAL BASIC

(i) Distinguish between Constant and Variable.

(ii) How variable is declared in Visual Basic?

(iii) Why is it important to assign a data type to a variable?

(iv) Define Control structure.

(v) Write down the name of any two control structure in Visual Basic.

(vi) Write down two uses of select case statement.

(vii) Why nested IF is complicated?

(viii) Define logical operator.

(ix) Give four examples of relational expression.

Q. SECTION-II

QNOTE. Attempt any one question. MS-ACCESS

Q5. (Marks 8) Define database model. Discuss different types of database models.

Q6. (Marks 8) What is field property in relation? Describe any six field properties in detail.

QNOTE. Attempt any two questions. Either from "C-Language" OR "Visual Basic"

Q. C-Language

Q7. (Marks 8) Define programming language. Describe any six characteristics of high level programming languages.

Q8. (Marks 8) A year is a leap year if it is divisible by 4, except that any year divisible by 100 is a leap year only if it is divisible by 400. Write a program that inputs a year such as 1996, 1800 and 2010, and displays "Leap year" if it is leap year, otherwise displays "Not Leap year".

Q9. (Marks 8) Write a program in C-Language that displays the sum and product of first ten even numbers.

Q. OR

Q. VISUAL BASIC

Q7. (Marks 8) Define programming language. Describe any six characteristics of high level programming languages.

Q8. (Marks 8) A year is a leap year if it is divisible by 4, except that any year divisible by 100 is a leap year only if it is divisible by 400, write a program that inputs a year such as 1996, 1800 and 2010, and displays "Leap year" if it is leap year, otherwise displays "Not Leap year".

Q9. (Marks 8) Write a program in Visual Basic that displays the sum and product of first ten even numbers.